Virtual BAS: a physics-grounded, real-world-like building automation system emulator with labelled faults, sensor imperfections and stochastic operation for developing and testing building AI

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Abstract

Artificial intelligence for buildings—fault detection and diagnosis, supervisory and reinforcement-learning control, and emerging *ambient* or large-language-model operators—must eventually work on the data of a real building automation system (BAS), where sensors drift, readings go missing, occupants behave randomly and equipment and control software fail. Real BAS data, however, are scarce and never come with ground truth, while simulation test beds are usually idealised. This paper presents *Virtual BAS*, an open-source emulator that couples a step-wise EnergyPlus simulation (the *inner space*, representing the physics) with an explicitly modelled *outer space* representing the real world: mathematically defined sensor and actuator models, BAS control sequences that act only on observed values, stochastic occupancy, weather and internal-gain events, and a library of 44 fault and event types on five layers (sensor, actuator, equipment, control, network). Every imperfection is labelled. A local-loop emulation by set-point translation lets sensor faults propagate physically even through loops that EnergyPlus solves internally. The simulation runs from real time to more than 92,000 times real time with interactive control of the clock, and faults can be injected while running by humans (web GUI), scripts, scenario files, a REST API, an MCP server for language-model agents and a Gymnasium interface. A generic BAS-ization procedure converts EnergyPlus models automatically; the DOE

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reference large office (1207 BAS points on 54 pieces of equipment) is the case study. Beyond the points of a typical BAS, the point list covers the full indoor environment (comfort indices, particles, volatile organic compounds, illuminance, sound, people counts and occupant feedback), an extended weather station with forecasts, sub-meters and grid signals, and every point is tagged as standard, advanced or next-generation instrumentation. With ideal sensors the emulator reproduces the native annual EnergyPlus results within 0.5% for electricity and 5.2% for natural gas (zone temperature mean absolute difference 0.03 K). Experiments demonstrate equipment-level fault impacts, labelled data generation for fault detection (detection macro-F1 0.74), AI agents injecting faults through the public API, the robustness gap of a supervisory AI controller between ideal and realistic outer spaces, the indoor-air-quality response to a wildfire-smoke episode that a standard BAS cannot observe, accurate time acceleration, and the automatic conversion of five reference buildings. Eighteen uses are identified, from AI development and benchmarking to university teaching, operator and facility-manager training, commissioning practice and the neutral evaluation of commercial analytics products.

Keywords: Building automation system, EnergyPlus co-simulation, Fault injection, Sensor drift, Labelled dataset, Fault detection and diagnosis, Reinforcement learning, Ambient AI, Digital twin, Large language model agents

1. Introduction

Commercial buildings are operated through building automation systems (BAS) that sample hundreds to thousands of points—temperatures, flows, positions, powers, statuses—and execute control sequences on them. These data are the substrate of almost every data-driven building application: automated fault detection and diagnosis (FDD) [1, 2], model predictive and reinforcement-learning (RL) control [3, 4, 5], and the recent wave of *ambient AI* agents and large language model (LLM) operators that are expected to monitor, explain and act on building operation continuously. The development of such methods faces a persistent dilemma. Real BAS data are difficult to obtain, are proprietary and building-specific, and above all have no ground truth: when a supply-air temperature reads 2 K above set point nobody knows whether the coil, the valve, the controller or the sensor is at

fault. Physics-based simulation provides ground truth, but simulation test beds are usually *idealised*: sensors are perfect, data are never missing, the controller sees the true state and nothing breaks. Methods developed in such environments routinely fail when transferred to real buildings, a gap that has been named as one of the central obstacles of RL in buildings [6, 7].

Building simulation test beds have made large progress. BCVTB [8] enabled co-simulation of EnergyPlus [9] with external controllers; the Modelica Buildings library [10] provides component-level dynamic models; BOPTEST [11] defines standardised emulators, key performance indicators and a REST interface for benchmarking control; the Advanced Controls Test Bed [12], Energym [13], Sinergym [14] and CityLearn [15] provide RL-oriented environments. For FDD, EnergyPlus has native fault models [16], fault modelling has been reviewed comprehensively [17], and labelled datasets with faulted and fault-free operation have been published [18, 19] together with a unified fault taxonomy [20]. What is still missing is an environment that behaves like a *real BAS as experienced by an AI*: one that (i) exposes the equipment-level point list of an actual BAS rather than a reduced observation vector, (ii) places a *real-world layer* between physics and observation in which sensor drift, noise, data loss and random occupancy are present by default, (iii) lets the BAS controllers act on the wrong values so that the physical consequences of imperfection are simulated rather than added afterwards, (iv) allows anyone—a person, a script, or an AI agent—to inject abnormal conditions *while the building runs*, at a controllable pace from real time to as fast as possible, and (v) labels everything with mathematically defined ground truth. The datasets cited above are static, and the control test beds are designed to benchmark controllers in idealised, fixed scenarios; to the author's knowledge none combines these five properties. A sixth requirement follows from field evidence on what BAS actually measure. In 90 buildings with semantic metadata, temperature streams outnumbered power meters by two orders of magnitude, and only 8% of the buildings had the points needed to compute an energy-use intensity [21]; installed CO_2 sensors had mean absolute errors of 118–138 ppm [22]. Indoor environmental quality beyond temperature, humidity and CO_2 [23], occupant counts and feedback, coil energy meters, actuator feedback and grid signals are rarely available. An emulator should therefore (vi) expose both the instrumentation of today's BAS and the next-generation points that AI applications need, so that the value of additional sensing can be quantified against ground truth.

The same gap affects people as much as algorithms. Operational problems

are pervasive and costly: commissioning of existing buildings, i.e. finding and fixing such problems, yielded a median whole-building energy saving of 16% in a compilation of 643 buildings [24], and correctly implemented and tuned controls, accurate sensors and fault correction could save about 29% of the energy of US commercial buildings [25]. Capturing these savings depends on operators, facility managers, technicians and commissioning agents who can read a BAS and diagnose what it shows. They learn mostly on live buildings, where faults cannot be staged on demand, the true state is unknown, and mistakes affect occupants and equipment. Aviation and nuclear power train and examine their operators on full-scope simulators governed by standards [26], and autonomous driving develops and validates its AI in open simulators before it reaches the road [27]. Building operation has only a few research prototypes of this kind, such as a hybrid simulator connected to a real controller for community-college training [28] and a video game that introduces a university HVAC-control course [29]. Ambient AI and LLM operators [30, 31], which are meant to run continuously beside or instead of a human operator, need such an environment even more: they must be trained, evaluated and stress-tested on realistic BAS behaviour before they are allowed to act on a real building. A realistic, interactive and labelled virtual BAS is therefore a shared infrastructure for AI research, education, workforce training and industry practice (Section 6.1).

This paper introduces *Virtual BAS*, an open-source framework that is built around a simple separation (Fig. 1). The *inner space* is an EnergyPlus model advanced step by step through the EnergyPlus Python API [32]; it represents the real physics. The *outer space* is the real world that surrounds the physics: sensors, actuators, the BAS software, occupants, weather perturbations, faults, operators and AI agents. A canonical example motivates the design: if a zone temperature sensor drifts, the VAV controller does not know it; it controls the drifting reading to set point, computes a wrong air-flow command and sends it to the physical model, so the true zone temperature departs from comfort while the BAS screen shows everything in order. In Virtual BAS this happens exactly so, and both trajectories and the defining equation of the drift are recorded.

The contributions are:

1. A formal inner/outer-space architecture for real-world-like BAS emulation, with mathematically defined sensor, actuator, occupancy and fault models (44 types on five layers) and a complete labelling scheme.

2. A *local-loop emulation by set-point translation* that propagates sensor errors physically through control loops that are solved inside EnergyPlus by ideal local controllers.
3. A generic *BAS-ization* procedure that converts EnergyPlus models into BAS-ready emulators: topology discovery, command binding, physical fault hooks, native control program extraction and semantic point list generation (Project Haystack [33] and Brick [34]) with an availability tier for every point (standard, advanced, next-generation).
4. An extended environment layer driven by EnergyPlus air flows and occupants—particle and VOC mass balances, illuminance, sound, occupant feedback, weather forecasts and grid signals—that completes the indoor and outdoor environment a BAS can observe.
5. An open-source implementation with a BAS-like web GUI, interactive time control (1× to as fast as possible), BACnet-like priority overrides and five interfaces for humans and AI agents (GUI, scenario files, REST, Model Context Protocol [35], Gymnasium [36]).
6. A case study on the DOE reference large office [37] with eight experiments: verification against native EnergyPlus, inner/outer coupling under sensor drift, an equipment-level fault impact matrix, labelled data generation and an FDD benchmark, AI agents in the loop, indoor air quality under wildfire smoke, time-acceleration accuracy, and generality across five reference buildings.
7. An analysis of eighteen uses of a virtual BAS across AI research, education and workforce training, industry practice and the research community, each mapped to the features that enable it and to the evidence in this paper.

The remainder is organised as follows. Section 2 reviews related work. Section 3 presents the methodology. Section 4 describes the case study and Section 5 the experiments. Section 6 discusses potential uses and limitations, and Section 7 concludes.

2. Related work

Control test beds. Co-simulation of building models with external controllers dates back to BCVTB [8]. BOPTEST [11] standardised the evaluation of advanced control with high-fidelity Modelica emulators [10], a REST API, baseline controllers and key performance indicators. The Advanced Controls

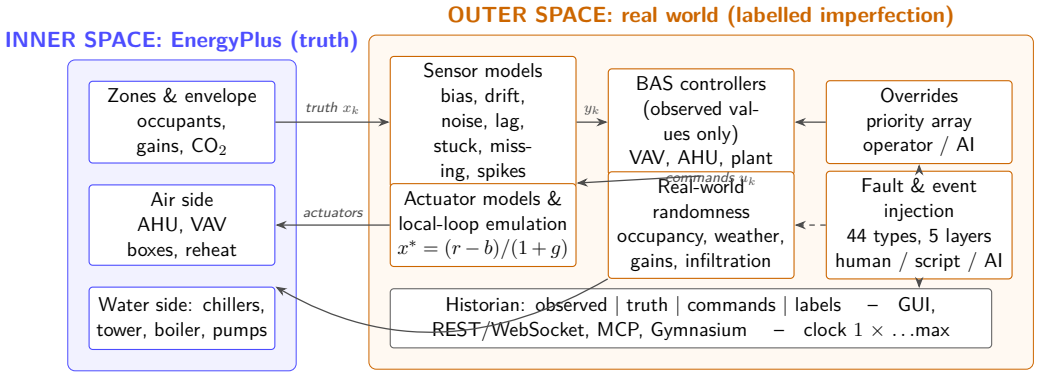


Figure 1: Inner/outer-space architecture of Virtual BAS. EnergyPlus (inner space) is advanced one BAS sample at a time; the outer space measures it through imperfect sensors, controls it through BAS sequences that only see observed values, and actuates it through actuator models. Faults and events modify sensors, controllers, actuators and the EnergyPlus physics; everything is recorded with ground-truth labels.

Test Bed [12] connects Spawn-of-EnergyPlus models to Python controllers through the BOPTEST and Alfalfa platforms. Energym [13], Sinergym [14] and CityLearn [15] provide Gym-style interfaces [38] for RL. These frameworks focus on *benchmarking controllers* in reproducible conditions. They typically expose a selected set of observations and control inputs, treat measurements as exact, and do not provide a library of injectable sensor, actuator, control and network faults during operation. Reviews of RL for buildings [3] and of real-world RL [6, 7] emphasise exactly the missing ingredients: partial and noisy observability, delays, safety constraints, non-stationarity and unexpected events.

Fault modelling and FDD data. Fault modelling for HVAC has been reviewed in [17]. EnergyPlus supports a set of operational faults (sensor offsets, fouling of coils, chillers, boilers, towers and air filters, thermostat offsets) [16]. ASHRAE RP-1312 [18] and the LBNL dataset [19] provide labelled faulted/fault-free data from simulation, laboratory and field, and a unified taxonomy organises HVAC faults [20]. These datasets are invaluable but *static*: an algorithm cannot interact with them, inject new faults, or close the loop. Classical FDD reviews [1, 2] repeatedly identify the lack of ground-truth data and of test environments with realistic sensor errors as a barrier to deployment.

Semantics and agents. Brick [34] and Project Haystack [33] provide semantic models for BAS points, enabling portable analytics. LLM agents interacting with tools through the Model Context Protocol [35] create a new class of building applications whose evaluation requires an interactive, semantically described BAS with ground truth.

Simulators for education and training. Full-scope training simulators are mandatory in nuclear power [26] and standard in aviation, and open simulators such as CARLA [27] underpin the development of autonomous-driving AI. In buildings, a hybrid simulator that connected a real-time HVAC simulation to a real controller supported a community-college curriculum on control configuration and troubleshooting [28], and a video game based on a building thermal model introduces a university course on HVAC control [29]. These tools teach control principles on simplified systems; none reproduces the point list, graphics, alarms and failure modes of a commercial BAS on a full building with labelled faults.

Gap. Virtual BAS is complementary to the above: it reuses the physics of EnergyPlus and the standard interfaces of RL environments, but its object of emulation is the *BAS as a whole*—point list, controllers, alarms, historian, operator overrides—embedded in a labelled real-world layer, interactive at any speed, open to fault injection by humans and AI, and obtained automatically from existing EnergyPlus models.

3. Methodology

3.1. Inner and outer space

Let $\mathbf{x}_k$ be the physical state of the building at BAS sample k as computed by EnergyPlus (the inner space), $\mathbf{y}_k$ the values reported by the BAS, $\mathbf{u}_k$ the controller outputs and $\mathbf{a}_k$ the values written to EnergyPlus actuators. With $\boldsymbol{\theta}(t)$ the parameters of the outer space, which are modified by injections, one BAS sample is

$$\mathbf{y}_k = \mathcal{S}\Big(\mathbf{x}_k; \boldsymbol{\theta}_S(t_k)\Big), \qquad \mathbf{u}_k = \mathcal{C}\Big(\hat{\mathbf{y}}_k; \boldsymbol{\theta}_C(t_k)\Big), \tag{1}$$

$$\mathbf{a}_k = \mathcal{A}\Big(\mathbf{u}_k, \mathbf{x}_k; \boldsymbol{\theta}_A(t_k)\Big), \qquad \mathbf{x}_{k+1} = \mathcal{E}\Big(\mathbf{x}_k, \mathbf{a}_k, \mathbf{w}_k; \boldsymbol{\theta}_E(t_k)\Big), \tag{2}$$

where $\mathcal{S}$ is the sensor layer, $\hat{\mathbf{y}}_k$ the BAS present values (last valid sample of $\mathbf{y}$), $\mathcal{C}$ the BAS control program, $\mathcal{A}$ the actuator layer including local-loop emulation, $\mathcal{E}$ one EnergyPlus time step with weather and stochastic

disturbances $\mathbf{w}_k$, and $\boldsymbol{\theta}_E$ the physical fault parameters. The controller never sees $\mathbf{x}_k$. All quantities $(\mathbf{x}_k, \mathbf{y}_k, \mathbf{u}_k, \boldsymbol{\theta}(t_k))$ are recorded.

3.2. Step-wise co-simulation and time control

EnergyPlus runs in a worker thread. At the beginning of every zone time step (callback *begin system time step before predictor*) the runner reads all output variables, meters and internal variables, calls the outer space and writes the returned actuator values. Supervisory plant on/off overrides, which EnergyPlus clears during its HVAC iterations, are re-applied in the *inside system iteration loop* callback, and weather perturbations are applied when EnergyPlus sets the current weather. EnergyPlus is suspended while the outer space computes, so the simulation is truly stepped. Sizing and warm-up periods are excluded from the BAS time line. Values read at the start of step k refer to the end of step $k-1$, which is the instant at which a BAS samples.

The pace is set by a speed factor s (simulated seconds per wall-clock second). With BAS interval Δt the target wall-clock time of sample k is $\tau_k = \tau_{k-1} + \Delta t/s$; the runner sleeps until τ_k in short slices so that speed changes, pause and single-step requests issued through the GUI or API take effect immediately. $s = \infty$ ("max") runs as fast as possible.

3.3. BAS-ization of an EnergyPlus model

A BAS-ready model is obtained from any EnergyPlus model by seven transformations (Table 1). The HVAC topology is discovered from EnergyPlus' own connection objects (`AirLoopHVAC`, branches, supply paths, zone splitters, `ZoneHVAC:EquipmentConnections`, plant and condenser loops), giving zones, terminal units, air handling units with their outdoor-air system, coils and fan, and plant loops with chillers, boilers, towers and pumps. Every quantity the BAS writes is bound to a dedicated `Schedule:Constant` ("command schedule"), e.g. a `SetpointManager:Scheduled` on each AHU supply-air node and plant supply node, the fan availability schedule, and the minimum-flow-fraction schedule of each VAV terminal (input method *Scheduled*). Because the minimum-flow input method changes EnergyPlus' autosizing of reheat coils, terminals and reheat coils are first hard-sized with the values of a sizing run of the native configuration. Physical faults are hooked through EnergyPlus `FaultModel:Fouling:*` objects whose availability and

severity schedules are again command schedules. Finally, the native schedules of fans, minimum outdoor air and thermostats are extracted so that the default BAS program reproduces the native operation of the source model.

Table 1: Generic transformations of the BAS-ization procedure.

T1	Simulation control: sizing performed, weather-file run period only
T2	Zone time step = BAS sampling interval (default 5 min)
T3	Run period and calendar year
T4	Remove file outputs (data are exchanged through the API)
T5	Command binding: SAT, plant supply temperatures, fan enable, VAV airflow commands → `Schedule:Constant`; terminals hard-sized from native sizing
T6	Physical fault hooks: chiller, boiler and tower fouling models with command schedules
T7	Zone CO_2 mass balance with actuatable outdoor concentration

From the topology a BAS point list is generated: for every piece of equipment the points a commercial BAS would expose (Table 2), each with a stable identifier (`AHU-2/SA-T`), unit, kind (sensor, setpoint, command, status, meter, virtual), sensor class, Project-Haystack style tags and a Brick class. Each sensor point is bound to an EnergyPlus output variable, meter or derived expression (e.g. valve position as water flow over design flow). Points that a BAS cannot measure but that are useful as labels (true occupant count, coil loads, COP, true outdoor-air fraction, infiltration) are recorded as *ground-truth-only* points. Duct static pressure, filter and loop differential pressures, which EnergyPlus' fan and pump models do not resolve, are provided by documented surrogate models, as are chiller refrigerant pressures (saturation temperatures from load- and fouling-dependent approach temperatures and an R-134a vapour-pressure fit), boiler flue temperature and fan and pump vibration.

Every point carries an *availability tier*. *Standard* points are those of the ASHRAE Guideline 36 point lists [39], vendor chiller integrations and utility meters; *advanced* points are found in well-instrumented BAS (actuator position feedback, airflow and water-flow meters, sub-meters, refrigerant-side chiller data, humidity and CO_2 in the return air); *next-generation* points are rarely installed today but valuable for analytics and AI (people counters, particle and VOC monitors, comfort stations, coil energy meters, vibration, weather forecasts, tariff, grid carbon intensity and demand-response signals,

occupant votes and complaints). BAS-calculated points (run-hours, start counts, kW/ton, tower approach, 15-min demand, degree-days, outdoor-air fraction from the mixing temperatures, Guideline 36 reset requests) are computed from *observed* values, so sensor errors propagate into them as in a real BAS. A scenario can restrict the exposed and recorded points to a set of tiers, e.g. to emulate a typical BAS of today while keeping the full ground truth.

3.4. Sensor model

For a physical quantity x_k the recorded value is

$$\tilde{x}_k = \tilde{x}_{k-1} + \frac{\Delta t}{\tau + \Delta t}\,(x_k - \tilde{x}_{k-1})\,, \tag{3}$$

$$d_k = d_{k-1} + r\,\Delta t + \vartheta(\mu_d - d_{k-1})\,\Delta t + \sigma_d\sqrt{\Delta t}\,\xi_k, \tag{4}$$

$$z_k = (1+g)\,\tilde{x}_k + b + d_k + \sigma_n\varepsilon_k + B_k\sigma_s\eta_k, \tag{5}$$

$$y_k = \mathrm{clip}(\delta\,\mathrm{round}(z_k/\delta),\, y_{\min},\, y_{\max})\,, \tag{6}$$

with first-order sensor dynamics (time constant τ), drift composed of a linear rate r and an Ornstein–Uhlenbeck component [40], installation gain error g and bias b, white noise, Bernoulli spikes $B_k \sim \mathrm{Bern}(p_s)$, quantisation δ and range limits. Two failure modes override the output: a stuck sensor returns the value at the stuck instant, and a missing sample is `NaN`. Missing data follow a two-state Gilbert–Elliott chain [41, 42] with transition probabilities p_{gb} and p_{bg}, producing realistic bursts with stationary loss rate $p_{gb}/(p_{gb} + p_{bg})$. Binary sensors (occupancy, status) use false-on/false-off probabilities. Installation parameters (b, g, r) are drawn once per sensor from data-sheet distributions of 36 sensor classes and scaled by a global *realism* factor (0 = ideal sensors, 1 = typical commercial hardware). Each sensor uses an independent random stream, so that scenarios with different faults share common random numbers. Fig. 2 illustrates the model.

3.5. Actuators and local-loop emulation

Explicit BAS outputs (VAV air-flow commands, outdoor-air damper position, fan start/stop) pass through actuator models that can be stuck at a position, failed, or overridden. Some loops, however, are closed *inside* EnergyPlus by ideal local controllers: the reheat valve to the zone heating set point, the AHU coil valves to the supply-air set point, chillers and boilers to their leaving water temperature, and the VAV box flow loop. A real local

Table 2: BAS points generated for each equipment type (abbreviated); S = standard, A = advanced, N = next-generation tier.

Zone	**S**: ZN-T, ZN-CO2, ZN-OCC (PIR), heating/cooling set points, occupant set-point adjust; **A**: ZN-RH, ZN-DP, illuminance, lighting level, window contact, lighting power; **N**: people count, mean radiant and operative temperature, PMV, PPD, PM2.5, PM10, TVOC, sound level, occupant votes and hot/cold complaints, plug power; truth: occupants, daylight, infiltration, solar and internal gains
VAV box	**S**: DA-F, DA-F-SP, DA-T, DMPR-CMD, loop outputs, terminal mode, G36 requests, low-flow alarm; **A**: DMPR-POS, RHV-POS, velocity pressure; **N**: reheat-coil water flow and energy meter; truth: RH-Q
AHU	**S**: SA-T, SA-F, MA-T, RA-T, OA-T, OA-F, set points, OAD-CMD, ECON-EN, fan command/status/speed/Hz/power, DSP, FLT-DP, freeze-stat, smoke detector, run-hours, starts, calculated OA fraction; **A**: SA-RH, SA-DP, RA-RH, RA-CO2, OA-RH, coil leaving-air temperatures, valve/damper feedback, fan current; **N**: coil water temperatures, flows and energy meters, coil pressure drop, fan vibration; truth: CC-Q, OA fraction
Chiller	**S**: KW, STS, CHW/CDW entering and leaving temperatures, %RLA, alarm, kW/ton, run-hours, starts; **A**: load, evaporator/condenser flows, refrigerant saturation temperatures and pressures; truth: PLR, COP
Boiler / tower / pump	**S**: STS, HW-LWT, HW-EWT, firing rate, alarm / fan power, status, speed, LWT, EWT, approach, range / KW, SPD, Hz, STS; **A**: load, gas, flow, flue temperature / make-up water; **N**: pump vibration
Plant loops	CHW-, HW-, CW-: ST, RT, F, DP, ST-SP, ΔT; chilled-water tons and plant kW/ton
Weather	**S**: OA-T, OA-RH, degree-days; **A**: wet bulb, dew point, enthalpy, humidity ratio, pressure, wind speed and direction, global irradiance, outdoor illuminance, rain; **N**: direct and diffuse irradiance, sky cover and temperature, precipitation, outdoor PM2.5/PM10, 1/6/24-h forecasts
Meters, grid	**S**: building electricity and gas demand, water, totalisers, 15-min and peak demand; **A**: end-use sub-meters (HVAC, lighting, cooling, fans, pumps, heat rejection, heating, exterior lighting, DHW), tower make-up water; **N**: plug and elevator sub-meters, tariff, carbon intensity, demand-response event, cost and carbon today

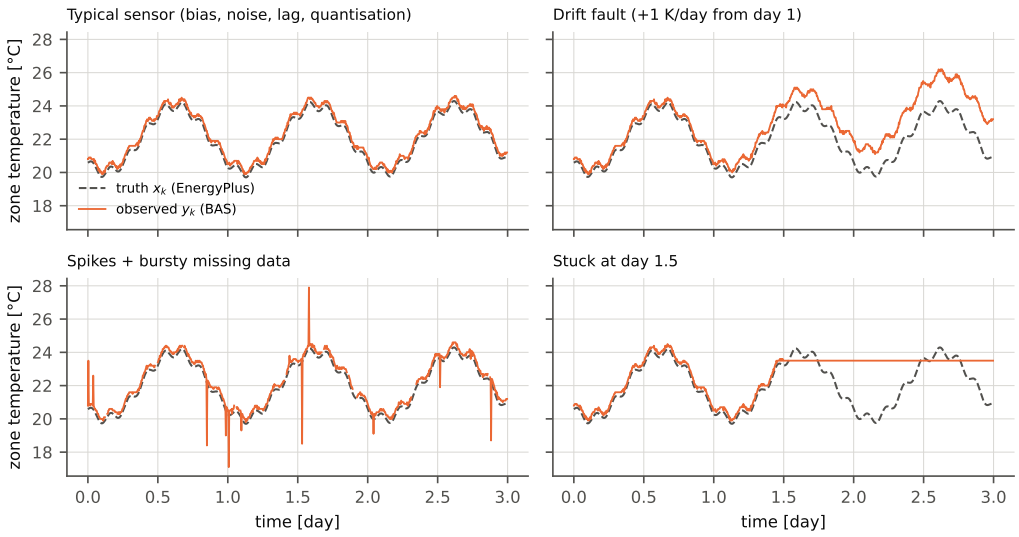


Figure 2: Sensor model output for a zone temperature sensor: typical commercial sensor, drift fault, spikes with bursty missing data, and stuck sensor (dashed: EnergyPlus truth).

loop drives its *measured* value to the set point r. With the deterministic sensor map $y = (1 + g)x + b$ (bias plus accumulated drift), the steady state of a loop with integral action satisfies $(1 + g)x^* + b = r$, hence

$$x^* = \frac{r - b}{1 + g}. \tag{7}$$

Virtual BAS therefore hands x^* instead of r to the ideal EnergyPlus controller (*set-point translation*). If the reading is stuck or spoofed at value v, the integral action winds up and the actuator saturates: $x^* = x_{\max}$ if $v < r$ and $x_{\min}$ otherwise for a direct process, and the opposite for a reverse (cooling) process. If the reading is unavailable, the loop holds its last output. With Eq. (7) every sensor fault propagates physically, including through loops that the BAS does not compute explicitly.

The VAV terminals are controlled explicitly: the BAS computes an air-flow set point, the box flow loop is emulated through Eq. (7) applied to the air-flow sensor, and the resulting flow fraction is written to the terminal's minimum-flow schedule while the EnergyPlus cooling set point of the zone is relaxed, so that EnergyPlus delivers exactly the commanded flow. Reheat is supplied by the reheat local loop. The outdoor-air flow is written to the

`Controller:OutdoorAir` actuator as the commanded outdoor-air fraction times the sum of commanded box flows.

3.6. Real-world randomness

Occupancy. Each zone with N design occupants follows the inhomogeneous Markov model of Page et al. [43], driven by the zone's EnergyPlus occupancy schedule $P(t)$ scaled by a daily attendance factor $a_d \sim \mathcal{U}(a_{\min}, 1)$:

$$T_{01} = \tfrac{\mu-1}{\mu+1} P_k + P_{k+1}, \qquad T_{11} = \tfrac{P_k-1}{P_k}\left(\tfrac{\mu-1}{\mu+1} P_k + P_{k+1}\right) + \tfrac{P_{k+1}}{P_k}, \tag{8}$$

$$n_{k+1} = \mathrm{Bin}(n_k, T_{11}) + \mathrm{Bin}(N - n_k, T_{01}), \tag{9}$$

with mobility μ. The count is written to the EnergyPlus *People* actuator, so occupants affect heat gains and CO_2; the BAS observes them only through PIR sensors with false-on/false-off rates (Fig. 6).

Events. Weather perturbations (heat waves, cold snaps, cloud cover) are applied to the EnergyPlus weather actuators as offsets/multipliers of the weather file; plug and lighting load changes, window opening (additional infiltration), meetings, holidays and outdoor CO_2 changes complete the event library.

3.7. Extended indoor and outdoor environment

EnergyPlus solves the heat, moisture and CO_2 balances. Quantities that it does not model, but that occupants experience and next-generation BAS measure, are computed in the outer space at every step from the EnergyPlus state, so they remain consistent with the simulated air flows, occupants, lighting power and solar gains.

Particles and VOC. Each zone is a well-mixed volume V supplied with Q_s by its terminal and Q_i by infiltration:

$$V\frac{dC}{dt} = Q_s C_s + P\, Q_i C_o + S - (Q_s + Q_i + kV)\, C, \tag{10}$$

$$C_s = (1-\eta)\left(x_{OA} C_o + (1 - x_{OA})\, \bar{C}_{ret}\right),$$

with outdoor concentration C_o, penetration factor P, deposition or sink rate k, indoor source S (occupants, materials, events), the AHU outdoor-air fraction x_{OA} taken from EnergyPlus, filter efficiency η by MERV class

[44] and the flow-weighted return concentration $\bar{C}_{ret}$ of the zones served by the AHU. Eq. (10) is integrated exactly over each step for PM2.5, the coarse fraction PM2.5–10 and TVOC (without filtration). Outdoor PM2.5 follows a log-normal Ornstein–Uhlenbeck process with a diurnal traffic cycle; `wildfire_smoke`, `indoor_pollutant_source` and `filter_bypass` injections act on C_o, S and η.

Light and sound. Work-plane illuminance combines the lumen method for electric light and the daylight-flux method, $E = E_d P_{light}/P_{light,d} + K_v U \Phi_{sol}/A$, with design illuminance E_d, luminous efficacy of daylight K_v, utilisation factor U and the solar flux Φ_{sol} transmitted by the zone's windows (EnergyPlus). The background sound level is the energetic sum of terminal noise following the fan law ($L_d + 50 \log_{10}(\dot{V}/\dot{V}_{max})$), speech scaled by the density of talking occupants, and a background floor.

Comfort and occupant feedback. Mean radiant and operative temperatures and the Fanger PMV and PPD [45] come from EnergyPlus and are exposed as comfort-station points. Occupants submit thermal-sensation votes $\mathrm{round}(\mathrm{PMV} + \sigma_i \varepsilon)$ through a feedback application at a Poisson rate, and file hot or cold complaints at a rate proportional to $N\,\mathrm{PPD}$, split by a logistic function of PMV; votes and complaint counts are BAS points.

Forecasts and grid. An hourly-issued 48-h weather forecast equals the weather file plus errors whose standard deviation grows with lead time, $\sigma(h) = \sigma_0 + \sigma_\infty(1 - e^{-h/\tau})$, and that are AR(1)-correlated between issues. The grid interface provides a time-of-use tariff, a synthetic marginal carbon intensity (diurnal profile, solar dip and AR noise) and demand-response events with a requested shed; cost and carbon are accumulated as KPIs. The forecast is also available as an API call and an MCP tool.

These models are deliberately simple, well documented and parameterised in the building configuration; they make the corresponding points behave plausibly and causally, not reproduce a particular building.

3.8. Faults and injections

All abnormal or random conditions are *injections* of a registered type. An injection has a target (point, equipment, zone or building), start t_0, optional

end t_1 and a severity profile

$$s(t) = \begin{cases} 0 & t < t_0 \text{ or } t \geq t_1 \\ 1 & \text{step} \\ \min\bigl(1, (t - t_0)/T_r\bigr) & \text{ramp (incipient)} \\ \mathbb{1}[((t - t_0) \bmod P) < D\,P] & \text{intermittent} \end{cases} \tag{11}$$

that modulates one parameter, $\theta(t) = \theta_0 + s(t)\,\Delta\theta$. Table 3 lists the library. Equipment faults act inside EnergyPlus: fouling factors of chillers, boilers and towers through the native fault models, filter loading through the fan pressure rise, fan degradation through the fan efficiency actuator, and trips and stuck-closed valves through plant supervisory overrides. New types are registered with a decorator and immediately appear in the GUI, the REST catalog and the MCP tools. A random fault generator draws arrivals from a Poisson process with rate λ per week and samples type, target, magnitude and duration from configurable distributions.

3.9. BAS control program

The BAS control program acts on present values only and comprises: occupancy scheduling and night-cycle operation; for each VAV box, cooling and heating PI loops on the zone temperature with air-flow set point between the minimum and maximum flow (optional dual maximum); for each AHU, supply fan start/stop, supply-air temperature set point, duct static pressure set point and an economizer with differential dry-bulb high limit, whose outdoor-air fraction combines a mixing-equation feed-forward on observed return and outdoor temperatures with a PI trim on the mixed-air temperature and a minimum outdoor-air limit computed from the sum of box air-flow set points; for the plant, chilled, hot and condenser water supply temperature set points. PI controllers are discrete with anti-windup and are tuned for the 5-min sampling interval. Two profiles are provided: *baseline*, which reproduces the native controls of the source model (including its schedules) and is used for verification, and *g36*, which implements trim-and-respond resets of supply-air temperature, duct static pressure and chilled-water temperature and outdoor-air reset of hot-water temperature inspired by ASHRAE Guideline 36 [39]. Any software point can be overridden through a BACnet-like 16-level priority array (operator default priority 8, AI agents 10, BAS program 16). A rule-based alarm engine raises alarms from observed values with on-delays, so that sensor faults can cause false alarms or mask real ones.

Table 3: Fault and event library (44 types). Each type has a defining equation, typed parameters and a target type; all are labelled with source (human, AI, scenario, random).

Layer	Target	Types (parameter)
Sensor	any sensor point	bias (b), drift (r, $d_{\max}$), gain (g), noise (σ), stuck, spikes (p, σ_s), dropout (p), override/spoofing (v)
Actuator	VAV, AHU	VAV damper stuck (position), reheat valve stuck closed/open, OA damper stuck, cooling/heating coil valve stuck closed, supply fan failure
Equipment	chiller, boiler, tower, AHU, pump	chiller/boiler/tower fouling (ratio r), filter fouling (Δp), fan degradation (ℓ), equipment trip
Control	AHU, VAV, zone	schedule error (always on/off), PI mistuning (gain multiplier), economizer disabled/always on, SAT reset disabled, controller frozen, set-point override
Network	any equipment, historian	communication loss (controller offline, BAS holds last values), historian outage
Events	zone, building	occupancy surge/vacancy, meeting, holiday, weather change (ΔT, $\Delta\phi$, solar), window open (ACH), plug and lighting load change, outdoor CO_2 change

3.10. Labels and data products

A run produces synchronised tables of observed values (the BAS trend log with gaps), truth, commands and per-sample label bit masks (missing, stuck, spike, sensor fault active, communication loss, false binary reading, spoofed), a long table of the severity of every active injection at every sample, the full injection definitions, sensor installation parameters, alarm history and metadata (scenario, seed, realism, control configuration, EnergyPlus version). The exact sensor error of every sample is $y_k - x_k$.

3.11. Interfaces for humans and AI

The web GUI (Fig. 3) mimics a commercial BAS front end with building overview, AHU, VAV and plant graphics, zone indoor-environment pages with a heat-map, a weather/forecast/grid/energy page, trends, alarms, a

point explorer with priority overrides and an injection console generated from the fault catalog. A global switch reveals the inner space (EnergyPlus truth) next to every observed value and highlights discrepancies. The same functions are available through a REST/WebSocket API with OpenAPI documentation, a dependency-free Python client, an MCP server exposing BAS tools (read points and trends, write set points, inspect the fault catalog, inject and clear faults, control the clock) to LLM agents [35], and an in-process Gymnasium environment [36] whose agent writes set points at priority 10 and observes BAS values, with rewards computed from the truth.

4. Case study

The DOE commercial reference large office (new construction, ASHRAE 90.1-2004, Chicago) [37] distributed with EnergyPlus 23.2 is used. The 12-storey, 46 320 m^2 building is modelled with 19 zones (basement, ground, a middle floor representing ten floors through a zone multiplier, and top floor, each with four perimeter and one core zone, plus plenums), four VAV air handling units with economizers, chilled- and hot-water coils and variable-speed fans, 16 VAV boxes with hot-water reheat, two water-cooled chillers, a cooling tower, a gas boiler and three pumps. BAS-ization with a 5-min sampling interval yields 1207 points on 54 pieces of equipment (722 sensor and meter points, 325 software points and 160 ground-truth-only points); 508 points are in the standard tier, 244 advanced and 295 next-generation, and 269 take their truth from outer-space or surrogate models. The model has 28 command schedules and 4 physical fault hooks. The Chicago TMY3 weather file is used and all runs use the calendar year 2023. Experiments were run on a Windows 11 laptop with EnergyPlus 23.2 and Python 3.12.

5. Experiments and results

5.1. Experiment 1: verification against native EnergyPlus

The emulator must not distort the physics. A full year was simulated (i) natively in EnergyPlus with the original controls and (ii) with Virtual BAS using ideal sensors (realism 0), deterministic occupancy and the baseline control profile, with identical time step. Table 4 and Fig. 4 compare hourly results. Annual whole-building electricity agrees within 0.5% (hourly CV(RMSE) 2.8%), cooling electricity within 1.3% and fan electricity within 0.2%; natural gas is 5.2% higher. The zone temperatures of the 16 zones

differ by 0.03 K on average (MAE of hourly means). Lighting and plug loads are identical by construction. Whole-building electricity and gas both satisfy the ASHRAE Guideline 14 hourly calibration criteria (NMBE ±10%, CV(RMSE) 30%) [46]; pump energy shows larger deviations. These residual differences are expected and have a clear origin: the native model uses ideal controllers solved within EnergyPlus' HVAC iterations, whereas the BAS controllers are sampled-data controllers acting every 5 min on values of the previous sample. The largest differences occur at morning start-up, when the economizer and the preheat coil interact during the first samples, and in pump energy, which reacts to short chilled-water demand events. Simulating a year took 300 s with Virtual BAS (105,120 BAS samples of 722 sensor points, about 351 samples per second) versus 35 s for the native batch simulation.

Table 4: Experiment 1: annual energy of native EnergyPlus and Virtual BAS (ideal sensors, baseline profile).

End use	Native [MWh]	Virtual BAS [MWh]	Diff. [%]	$\mathrm{CV(RMSE)}_h$ [%]
Electricity, total	5,729	5,756	+0.5	2.8
Natural gas, total	1,580	1,663	+5.2	15.0
Cooling (chillers)	509	502	-1.3	13.9
Fans	178	178	+0.2	5.8
Pumps	258	295	+14.1	44.4
Heat rejection	139	136	-2.3	20.7
Heating (boiler)	1,511	1,594	+5.5	15.7
Interior lighting	1,566	1,566	-0.0	0.0
Interior equipment	2,329	2,329	+0.0	0.0
Zone temp. MAE / RMSE [K]	0.029 / 0.091 (mean of 16 zones)			

5.2. Experiment 2: inner/outer-space coupling under sensor drift

The motivating example was simulated for one summer week with realistic sensors and stochastic occupancy: the zone temperature sensor of the core zone of the middle floors (representing ten floors) drifts by +0.5 K/day from the second day, compared with a fault-free twin run with identical random streams (Fig. 5). The VAV controller keeps the *observed* temperature at the cooling set point throughout. The *true* temperature, however, decreases day by day; at the end of the week the sensor reads 2.8 K above the truth. The controller compensates the apparent overheating with more air: the mean discharge air flow during occupied hours increases by 15%, and the building's HVAC electricity increases by 4.2%. None of this is visible to a

BAS operator looking at the zone temperature trend. The run demonstrates the central mechanism of Virtual BAS: an outer-space imperfection produces a wrong control signal, the wrong signal changes the physics, and the change is labelled.

5.3. Experiment 3: equipment-level fault impact matrix

Twenty-seven single-fault scenarios covering all layers were each simulated for a summer and a winter week with realistic sensors and stochastic occupancy and compared with the fault-free twin (common random numbers). Fig. 7 reports the changes in electricity, gas and discomfort (occupied K·h outside the set-point band ±0.5 K, computed from the truth); rows are grouped by layer (sensor, actuator, equipment, control, network, events). The matrix shows physically consistent, season-dependent behaviour at equipment level. A stuck-open outdoor-air damper of the main AHU increases winter gas by 157% but has little effect in summer; a closed damper saves heating energy at the cost of ventilation. A leaking reheat valve increases summer gas consumption 16-fold relative to the fault-free week and causes 114 K·h of overheating. A stuck-closed chilled-water valve of AHU-1 produces 1210 K·h of summer discomfort but no winter effect; a stuck-closed reheat valve does the opposite (162 K·h in winter). Chiller and tower fouling raise summer electricity by 1.0% and 3.3%, boiler fouling raises winter gas by 55%, filter fouling and fan degradation raise fan energy. A trip of one of the two chillers *reduces* summer electricity because the remaining chiller operates at a more efficient part-load ratio, while discomfort increases at peak load—a reminder that fault impacts are not always energy penalties. A 24/7 schedule error on the main AHU increases summer electricity by 14.4%. Several faults are *silent* in one season (e.g. economizer and mixed-air sensor faults in summer, stuck-closed reheat valve in summer), an important property of real fault data that a dataset generator must reproduce: labels mark fault presence, while the twin comparison quantifies fault impact.

5.4. Experiment 4: labelled data generation and an FDD benchmark

Two independent datasets were generated with the random fault generator restricted to nine AHU-level fault types (sensor bias, drift and stuck on AHU temperature sensors, stuck outdoor-air damper, stuck-closed chilled-water valve, filter fouling, fan degradation, disabled economizer, schedule error) at $\lambda = 3$ per week: a training set (April–June) and a test set (July–August) with different seeds. Hourly features (means and standard deviations

of 23 quantities, e.g. supply-air temperature error, mixing-equation residual, outdoor-air damper command/feedback deviation, fan power per flow, filter pressure drop) were computed from the *observed* trend logs only, and labels from the injection severity tables. A random forest [47, 48] trained on 8,064 AHU-hours reaches a detection macro-F1 of 0.74 and a diagnosis macro-F1 of 0.24 on 5,376 test AHU-hours (Fig. 8). The difficulty is genuine and informative: the test period lies in a different season, background sensor errors and missing data are present, and several faults are unobservable under the prevailing conditions (e.g. a disabled economizer in hot weather). The experiment demonstrates that Virtual BAS produces ML-ready, consistently labelled data with controllable difficulty, distribution shift and realistic class imbalance.

5.5. Experiment 5: AI agents in the loop

AI fault injection through the public API. An autonomous adversarial agent operated the building for six days through the REST API, in lock-step with the simulation (pause, 6 h steps). Every six simulated hours it read the BAS like an operator and chose an injection: forcing 100% outdoor air on the AHU with the largest fan power when the outdoor air was hot, an after-hours schedule error in the evening, and otherwise a slow drift of a zone thermostat. It injected 24 faults, all recorded with `source=ai` and the agent's rationale. The same calls are exposed as MCP tools, so an LLM agent can play the same role, for example to stress-test an FDD system.

Robustness of a supervisory AI controller. A supervisory agent resetting the AHU supply-air and chilled-water set points from the observed outdoor-air temperature was evaluated through the Gymnasium interface in three outer spaces—ideal sensors, realistic sensors, and realistic sensors with a +4 K bias of the outdoor-air sensor—for two weeks in the shoulder season and in summer, each against the baseline BAS program in the same outer space (Fig. 9). In the shoulder season the agent reduces electricity by 5.2% with ideal sensors and 6.4% with realistic sensors, but only by 3.3% when the outdoor-air sensor reads 4 K high, because the biased reading shifts every reset decision. At the same time the warmer supply air increases preheat and reheat gas use by 21 MWh in two weeks (realistic sensors), so that site energy changes by +2.5%; in summer the electricity saving is small (0.3%). An evaluation restricted to electricity and to ideal sensors would therefore have overstated the value of this policy twice. Such trade-offs and their

sensitivity to measurement errors must be understood before an AI controller is deployed, and they cannot be studied in an idealised environment.

5.6. Experiment 6: indoor environment under wildfire smoke

A summer week with a 30-h wildfire-smoke episode (outdoor PM2.5 up to 133 $\mu g/m^3$) was simulated with the reference MERV 8 filters, with MERV 13 filters, with a filter-bypass fault (60% bypass, no pressure-drop symptom) and with an IEQ-aware sequence that locks the economizers out while the outdoor PM2.5 exceeds 35 $\mu g/m^3$ (Fig. 10a). Because the economizers bring up to 100% outdoor air in this season, the mean indoor concentration peaks at 93 $\mu g/m^3$ with MERV 8, 111 $\mu g/m^3$ with the bypass fault, 66 $\mu g/m^3$ with the economizer lock-out and 45 $\mu g/m^3$ with MERV 13; occupied zone-hours above 35 $\mu g/m^3$ are 362, 367, 346 and 104. None of this is visible in the standard tier: a typical BAS of today has no particle sensor, the bypass fault has no pressure-drop symptom, and the lock-out decision requires the next-generation outdoor PM2.5 point. Figs. 10b–d show one day of the other environment points of a perimeter zone: illuminance with its daylight share and the photosensor reading, TVOC flushed by the morning ventilation and its noisy metal-oxide reading, sound level, and PMV with the sparse mean votes of the zone's occupants.

5.7. Experiment 7: time acceleration

The pacing accuracy was measured by setting target speed factors while running (Fig. 11). The achieved speed matches the target within 0.00% for factors from 10 to 3600; at maximum speed one BAS sample takes 3.3 ms, i.e. 92,000 times real time, so one simulated year runs in about 5.0 minutes. Real time (1×) follows from the same pacing law. The speed can be changed, paused or single-stepped at any moment from the GUI or the API, which allows the same environment to serve real-time interaction with people or deployed agents and fast training of learning agents.

5.8. Experiment 8: generality

The BAS-ization procedure was applied without any building-specific code to five DOE reference buildings (Table 5). All models were converted and simulated for one week. Multi-zone VAV systems with water-based plants (large office, hospital, secondary school) are fully BAS-controlled; the medium office (packaged VAV with DX cooling and electric reheat) and the small office (packaged single-zone units) are monitored completely and their

terminals and thermostats are BAS-controlled while unsupported components keep their native local control. The hospital, with 55 zones and four 100%-outdoor-air and mixed-air AHUs, yields 3244 points on 128 pieces of equipment and runs at 60 samples per second.

Table 5: Experiment 8: automatic BAS-ization of five DOE reference buildings (one-week runs).

Building	Zones	VAV	AHUs	Plant loops	Equipment	Points	Samples/s
Large office	16	16	4	4	54	1207	141
Medium office	15	15	3	1	41	906	227
Small office	5	0	5	1	15	425	358
Hospital	55	55	4	4	128	3244	60
Secondary school	46	41	9	3	107	2828	70

6. Discussion

6.1. Potential uses and impact

A real BAS is a poor place to learn, experiment or certify: faults cannot be staged on demand, the true state is unknown, every mistake affects occupants and equipment, a year of operation takes a year, and access is restricted by owners, vendors and cyber-security policies. Virtual BAS removes each of these barriers at once—it is safe to break, labelled, repeatable through seeded scenarios, faster than real time, free and open, and it can be generated for any EnergyPlus model. Table 6 lists eighteen uses that follow from these properties, grouped into four areas, together with the users, the enabling features and the evidence available in this paper. The table distinguishes uses that the experiments demonstrate from uses for which the interfaces exist but no evaluation is reported here.

Research and development of building AI. Virtual BAS is first a proving ground for building AI. FDD developers obtain unlimited labelled data with controllable fault rates, severities, seasons and sensor quality and can test algorithms in closed loop (Experiments 3 and 4). Control and RL researchers can evaluate policies under partial observability, sensor errors, missing data and equipment faults, and quantify robustness by switching the outer space while keeping the inner space fixed (Experiment 5). Ambient-AI and LLM agents can be connected through MCP and evaluated on operational tasks

Table 6: Potential uses of Virtual BAS, with primary users in italics. Evidence: Exp. n = demonstrated in Experiment n; *interf.* = all required functions are implemented but not evaluated in this paper; *partial* = some functions are missing (see Section 6.2).

Use (*users*)	What Virtual BAS provides	Evid.
Research and development of building AI		
FDD development and benchmarking (*FDD researchers, analytics developers*)	Unlimited labelled data with controlled fault type, severity, season and sensor quality; closed-loop tests	Exp. 3, 4
Supervisory, MPC and RL control (*control and RL researchers*)	Gymnasium; sensor errors, missing data and faults; ideal vs. realistic outer space on the same physics	Exp. 5
Ambient-AI and LLM operators (*AI researchers, BAS vendors*)	MCP tools, overrides and alarms at any speed; ground truth for scoring; agents as operators and adversaries	Exp. 5
Foundation models (*machine-learning researchers*)	Diverse, labelled, multivariate BAS time series for any EnergyPlus model [49]	Exp. 4, 8
Virtual sensing and sensor recalibration (*researchers*)	Observed and true values side by side; defined bias, drift, lag and data-loss models	Exp. 2
Metadata inference, portable apps (*researchers, integrators*)	Haystack/Brick point lists with known metadata for converted buildings [50]	Exp. 8
Value of information, sensor design (*sensor makers, standard writers*)	Standard, advanced and next-generation tiers; one scenario seen through different instrumentation	Exp. 6
Education and workforce development		
University courses (*instructors, students*)	BAS-like GUI; equipment that can be broken safely; truth switch; open scenarios; a season in one lab session	interf.
Operator and facility-manager training (*owners, utilities, training programmes*)	Drills with hidden or random faults; alarms, trends and overrides as in a real BAS; labels as answer key	interf.
Commissioning practice (*commissioning agents*)	Functional tests of sequences, staged faults, trend review against known causes	interf.
Emergency and resilience drills (*facility managers*)	Wildfire smoke, heat waves, cold snaps, equipment trips, demand response at run time	Exp. 6
Alarm management, HMI research (*human-factors researchers*)	Reproducible alarm floods and faults for user studies under identical conditions [51]	interf.
Industry practice		
Testing commercial FDD and analytics (*owners, efficiency programmes*)	Blind test buildings with known faults and severities; scoring against labels	Exp. 4
Sequence verification, virtual commissioning (*designers, contractors*)	G36-style sequences in closed loop and under sensor faults before site work [52]	Exp. 2, 3
BAS and analytics software (*vendors, start-ups*)	A mock building behind REST, WebSocket and MCP for integration tests and demonstrations	interf.
Cyber-physical security (*security researchers*)	Spoofed readings, malicious overrides, controller and historian outages, physical impact [53]	partial
Grid-interactive buildings (*utilities, aggregators*)	Tariff, carbon intensity, demand-response events, demand meters [54]	interf.
Research community		
Benchmarks, competitions, reproducibility (*all*)	Seeded scenarios, open building models, common KPIs, public code	Exp. 1–8

(diagnosing complaints, finding faults, writing safe overrides) with exact ground truth; they can also act as adversaries that generate challenging scenarios. Because any EnergyPlus model can be converted, the emulator can also produce the large and diverse corpora of BAS-level, multivariate, labelled time series that foundation models for building operation will need; existing synthetic corpora for pre-training cover whole-building loads only [49]. The point tiers support sensor-placement and value-of-information studies: the same scenario can be observed through the instrumentation of a typical BAS and through next-generation sensing, and the benefit of, e.g., coil energy meters, people counters or particle monitors for a given task can be measured against ground truth (Experiment 6).

Education and workforce development. Students rarely get write access to a real BAS, and when they do they cannot break it. With Virtual BAS a course on HVAC systems, controls or building data science can let every student operate the same building, change set points and sequences, inject a stuck damper or a drifting sensor and see both what the BAS shows and what really happens; time acceleration fits a season of operation into a laboratory session. For operators, facility managers and technicians the same features become troubleshooting drills. The GUI hides the ground truth by default, an instructor injects faults from the console or lets the random fault generator choose them, the trainee works only with graphics, trends, alarms and overrides, and the label record then serves as the answer key for scoring. This brings to building operation the simulator-based training that is standard in aviation and nuclear power [26] and has so far been limited to prototypes in buildings [28, 29]. Commissioning agents can rehearse functional tests, and facility managers can practise responses to wildfire smoke, heat waves, equipment trips and demand-response events. Reproducible alarm floods also make it possible to study alarm management and operator interfaces with human participants [51].

Industry practice. Owners and efficiency programmes that procure FDD or analytics services currently have no neutral way to compare products; a blind Virtual BAS building with known faults gives such a test, scored against the labels (Experiment 4). Designers and control contractors can exercise sequences in closed loop, including their behaviour under sensor faults, before the site work, complementing formal specification and verification workflows [52]. BAS and analytics vendors obtain a mock building for continuous integration, demonstrations and training of their own staff. Security researchers

can emulate spoofed readings with sensor-layer faults, malicious commands with overrides and outages with network-layer faults, and observe the physical impact of an attack [53]. Utilities and aggregators can test demand-response and carbon-aware strategies with the tariff, grid and demand-meter points [54]. Given the savings at stake in commissioning [24] and controls [25], even a modest improvement in how people and software operate buildings has a large aggregate effect.

Research community. Seeded scenarios, open building models and common KPIs allow benchmarks and competitions in which every participant faces the same faults, sensor errors and occupants, and allow results to be reproduced exactly. This addresses a recurrent obstacle of FDD and building-AI research, in which methods are evaluated on private data that others cannot access [1, 2].

6.2. Limitations and future work

Limitations. (i) Local loops inside EnergyPlus are ideal; set-point translation reproduces their steady state under sensor error but not their fast dynamics, hunting or valve characteristics. (ii) Duct static and loop differential pressures are surrogate models because EnergyPlus does not resolve pressure networks. (iii) The 5-min sampling interval limits the dynamics of BAS loops; finer intervals are possible at proportionally higher cost. (iv) Explicit BAS control currently covers multi-zone VAV systems with water-based plants; other system types are monitored and keep native control. (v) Fault parameters follow data-sheet-type distributions rather than calibrated field statistics; the framework accepts any distribution. (vi) The verification shows residual differences of a few percent in gas and pump energy caused by sampled-data versus ideal control. (vii) The environment does not yet speak building network protocols. (viii) The extended environment models (well-mixed zones, lumen and daylight-flux methods, fan-law acoustics, stochastic feedback, synthetic grid signals) and the refrigerant, flue and vibration surrogates are simplified and not calibrated against measurements; the filter pressure drop does not yet depend on the MERV class, and ozone, NO_2, formaldehyde, glare and envelope heat flux are not modelled. The tier assignment reflects guideline point lists and field surveys and will differ between buildings.

Future work. A BACnet/IP and Modbus gateway so that commercial BAS front ends and analytics can connect unchanged; Spawn-of-EnergyPlus or

Modelica inner spaces for faster local dynamics; calibrated fault occurrence statistics; multi-building portfolios; and standardised benchmark tasks for LLM-based building operators. For education and training, an instructor/trainee mode that withholds labels and active injections from the trainee's session and scores diagnoses automatically, together with course modules and drill libraries developed and evaluated with instructors and building operators. For security studies, a dedicated library of cyber-attacks on BAS protocols. The educational and training uses in Table 6 have not yet been evaluated with learners; such studies are planned.

7. Conclusions

This paper presented Virtual BAS, a real-world-like building automation system emulator that separates the physics (EnergyPlus, inner space) from a mathematically defined and fully labelled real world (outer space). Sensors drift, data go missing, occupants move randomly, equipment and control software fail, and the BAS acts on what it sees, so the consequences of imperfection are simulated rather than superimposed. Faults and events can be injected at run time by humans and AI agents, the clock runs from real time to more than 92,000 times real time, and any EnergyPlus model can be converted. For the DOE large office, the emulator reproduces native EnergyPlus within 0.5% in annual electricity with ideal sensors, produces physically consistent equipment-level fault signatures, generates labelled datasets for FDD, supports AI agents as operators, controllers and adversaries, and exposes robustness gaps that idealised environments hide. Beyond AI research, the same properties—safe to break, labelled, repeatable, fast and open—make it a training simulator for students, operators, facility managers and commissioning agents, a neutral test building for commercial FDD and analytics products, and a shared benchmark for the research community. The software, building models, scenarios, experiments and tutorials are openly available to help the community develop building AI, and the people who operate buildings, for real buildings.

CRediT authorship contribution statement

Liang Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The software, the prepared building models, all scenarios, the experiment scripts and tutorials are openly available at `https://github.com/TensorBuildLab/Virtual-BAS` under the MIT license. All results in this paper can be regenerated with the scripts in the `experiments/` and `paper/` folders.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist with software implementation, running the experiments and drafting the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

[1] S. Katipamula, M. R. Brambley, Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I, HVAC&R Research 11 (1) (2005) 3–25.

[2] W. Kim, S. Katipamula, A review of fault detection and diagnostics methods for building systems, Science and Technology for the Built Environment 24 (1) (2018) 3–21.

[3] Z. Wang, T. Hong, Reinforcement learning for building controls: The opportunities and challenges, Applied Energy 269 (2020) 115036. `doi:10.1016/j.apenergy.2020.115036`.

[4] T. Wei, Y. Wang, Q. Zhu, Deep reinforcement learning for building HVAC control, in: Proceedings of the 54th Annual Design Automation Conference (DAC '17), 2017, pp. 1–6.

[5] Z. Zhang, K. P. Lam, Practical implementation and evaluation of deep reinforcement learning control for a radiant heating system, in: Proceedings of the 5th Conference on Systems for Built Environments (BuildSys '18), 2018, pp. 148–157.

[6] K. Nweye, B. Liu, P. Stone, Z. Nagy, Real-world challenges for multi-agent reinforcement learning in grid-interactive buildings, Energy and AI 10 (2022) 100202. doi:10.1016/j.egyai.2022.100202.

[7] G. Dulac-Arnold, N. Levine, D. J. Mankowitz, J. Li, C. Paduraru, S. Gowal, T. Hester, Challenges of real-world reinforcement learning: definitions, benchmarks and analysis, Machine Learning 110 (9) (2021) 2419–2468.

[8] M. Wetter, Co-simulation of building energy and control systems with the Building Controls Virtual Test Bed, Journal of Building Performance Simulation 4 (3) (2011) 185–203. doi:10.1080/19401493.2010.518631.

[9] D. B. Crawley, L. K. Lawrie, F. C. Winkelmann, W. F. Buhl, Y. J. Huang, C. O. Pedersen, R. K. Strand, R. J. Liesen, D. E. Fisher, M. J. Witte, J. Glazer, Energyplus: creating a new-generation building energy simulation program, Energy and Buildings 33 (4) (2001) 319–331. doi:10.1016/S0378-7788(00)00114-6.

[10] M. Wetter, W. Zuo, T. S. Nouidui, X. Pang, Modelica Buildings library, Journal of Building Performance Simulation 7 (4) (2014) 253–270. doi:10.1080/19401493.2013.765506.

[11] D. Blum, J. Arroyo, S. Huang, J. Drgoňa, F. Jorissen, H. T. Walnum, Y. Chen, K. Benne, D. Vrabie, M. Wetter, L. Helsen, Building optimization testing framework (BOPTEST) for simulation-based benchmarking of control strategies in buildings, Journal of Building Performance Simulation 14 (5) (2021) 586–610. doi:10.1080/19401493.2021.1986574.

[12] T. Marzullo, S. Dey, N. Long, J. Leiva Vilaplana, G. Henze, A high-fidelity building performance simulation test bed for the development and evaluation of advanced controls, Journal of Building Performance Simulation 15 (3) (2022) 379–397. doi:10.1080/19401493.2022.2058091.

[13] P. Scharnhorst, B. Schubnel, C. Fernández Bandera, J. Salom, P. Taddeo, M. Boegli, T. Gorecki, Y. Stauffer, A. Peppas, C. Politi, Energym: A building model library for controller benchmarking, Applied Sciences 11 (8) (2021) 3518. doi:10.3390/app11083518.

[14] J. Jiménez-Raboso, A. Campoy-Nieves, A. Manjavacas-Lucas, J. Gómez-Romero, M. Molina-Solana, Sinergym: a building simulation and control framework for training reinforcement learning agents, in: Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '21), 2021, pp. 319–323.

[15] J. R. Vázquez-Canteli, J. Kämpf, G. Henze, Z. Nagy, CityLearn v1.0: An OpenAI Gym environment for demand response with deep reinforcement learning, in: Proceedings of the 6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '19), 2019, pp. 356–357.

[16] R. Zhang, T. Hong, Modeling of HVAC operational faults in building performance simulation, Applied Energy 202 (2017) 178–188.

[17] Y. Li, Z. O'Neill, A critical review of fault modeling of HVAC systems in buildings, Building Simulation 11 (5) (2018) 953–975. doi:10.1007/s12273-018-0458-4.

[18] J. Wen, S. Li, RP-1312 – tools for evaluating fault detection and diagnostic methods for air-handling units, Tech. rep., ASHRAE, Atlanta, GA (2012).

[19] J. Granderson, G. Lin, Y. Chen, A. Casillas, J. Wen, Z. Chen, P. Im, S. Huang, J. Ling, A labeled dataset for building HVAC systems operating in faulted and fault-free states, Scientific Data 10 (2023) 342. doi:10.1038/s41597-023-02197-w.

[20] Y. Chen, G. Lin, E. Crowe, J. Granderson, Development of a unified taxonomy for HVAC system faults, Energies 14 (17) (2021) 5581. doi:10.3390/en14175581.

[21] G. Fierro, M. Pritoni, M. AbdelBaky, P. Raftery, T. Peffer, G. Thomson, D. E. Culler, Mortar: An open testbed for portable building analytics, ACM Transactions on Sensor Networks 16 (1) (2019).

[22] W. J. Fisk, D. P. Sullivan, D. Faulkner, E. Eliseeva, CO_2 monitoring for demand controlled ventilation in commercial buildings, Tech. Rep. LBNL-3279E, Lawrence Berkeley National Laboratory (2010).

[23] International WELL Building Institute, WELL building standard, version 2, https://v2.wellcertified.com (2020).

[24] E. Mills, Building commissioning: a golden opportunity for reducing energy costs and greenhouse gas emissions in the United States, Energy Efficiency 4 (2) (2011) 145–173. doi:10.1007/s12053-011-9116-8.

[25] N. Fernandez, S. Katipamula, W. Wang, Y. Xie, M. Zhao, C. Corbin, Impacts of commercial building controls on energy savings and peak load reduction, Tech. Rep. PNNL-25985, Pacific Northwest National Laboratory, Richland, WA (2017).

[26] American Nuclear Society, ANSI/ANS-3.5-2018 (R2024): Nuclear power plant simulators for use in operator training and examination, American Nuclear Society, La Grange Park, IL (2018).

[27] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, V. Koltun, CARLA: An open urban driving simulator, in: Proceedings of the 1st Annual Conference on Robot Learning (CoRL), Vol. 78 of Proceedings of Machine Learning Research, 2017, pp. 1–16.

[28] P. Xu, P. Haves, J. J. Deringer, A simulation-based testing and training environment for building controls, in: Proceedings of SimBuild 2004, IBPSA-USA National Conference, Boulder, CO, 2004.

[29] H. Johra, L. Rohde, E. Petrova, Video game-based learning for teaching building thermodynamics and control of HVAC systems, in: Proceedings of BuildSim-Nordic 2020, Oslo, Norway, 2020.

[30] L. Zhang, Z. Chen, V. Ford, Advancing building energy modeling with large language models: Exploration and case studies, Energy and Buildings 323 (2024) 114788. doi:10.1016/j.enbuild.2024.114788.

[31] L. Zhang, Z. Chen, Large language model-based interpretable machine learning control in building energy systems, Energy and Buildings 313 (2024) 114278. doi:10.1016/j.enbuild.2024.114278.

[32] U.S. Department of Energy, EnergyPlus Python API documentation (version 23.2), https://energyplus.readthedocs.io/en/latest/api.html, accessed 8 November 2026 (2023).

[33] Project Haystack, Project Haystack: open source initiative to streamline working with data from the internet of things, https://project-haystack.org, accessed 8 November 2026 (2026).

[34] B. Balaji, A. Bhattacharya, G. Fierro, J. Gao, J. Gluck, D. Hong, A. Johansen, J. Koh, J. Ploennigs, Y. Agarwal, M. Bérges, D. Culler, R. Gupta, M. B. Kjærgaard, M. Srivastava, K. Whitehouse, Brick: Towards a unified metadata schema for buildings, in: Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys '16), 2016, pp. 41–50. doi:10.1145/2993422.2993577.

[35] Anthropic, Model Context Protocol specification, https://modelcontextprotocol.io, accessed 8 November 2026 (2024).

[36] M. Towers, A. Kwiatkowski, J. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão, A. Kallinteris, M. Krimmel, A. KG, et al., Gymnasium: A standard interface for reinforcement learning environments, arXiv:2407.17032 (2024).

[37] M. Deru, K. Field, D. Studer, K. Benne, B. Griffith, P. Torcellini, B. Liu, M. Halverson, D. Winiarski, M. Rosenberg, M. Yazdanian, J. Huang, D. Crawley, U.s. department of energy commercial reference building models of the national building stock, Tech. Rep. NREL/TP-5500-46861, National Renewable Energy Laboratory, Golden, CO (2011).

[38] G. Brockman, V. Cheung, L. Pettersson, J. Schneider, J. Schulman, J. Tang, W. Zaremba, OpenAI Gym, arXiv:1606.01540 (2016).

[39] ASHRAE, ASHRAE Guideline 36-2021: High-Performance Sequences of Operation for HVAC Systems, ASHRAE, Atlanta, GA, 2021.

[40] G. E. Uhlenbeck, L. S. Ornstein, On the theory of the Brownian motion, Physical Review 36 (5) (1930) 823–841. doi:10.1103/PhysRev.36.823.

[41] E. N. Gilbert, Capacity of a burst-noise channel, Bell System Technical Journal 39 (5) (1960) 1253–1265. doi:10.1002/j.1538-7305.1960.tb03959.x.

[42] E. O. Elliott, Estimates of error rates for codes on burst-noise channels, Bell System Technical Journal 42 (5) (1963) 1977–1997. doi:10.1002/j.1538-7305.1963.tb00955.x.

[43] J. Page, D. Robinson, N. Morel, J.-L. Scartezzini, A generalised stochastic model for the simulation of occupant presence, Energy and Buildings 40 (2) (2008) 83–98.

[44] ASHRAE, ANSI/ASHRAE standard 52.2-2017: Method of testing general ventilation air-cleaning devices for removal efficiency by particle size, ASHRAE, Atlanta (2017).

[45] ISO, ISO 7730:2005 ergonomics of the thermal environment – analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria, International Organization for Standardization, Geneva (2005).

[46] ASHRAE, ASHRAE Guideline 14-2014: Measurement of Energy, Demand, and Water Savings, ASHRAE, Atlanta, GA, 2014.

[47] L. Breiman, Random forests, Machine Learning 45 (1) (2001) 5–32. doi:10.1023/A:1010933404324.

[48] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in Python, Journal of Machine Learning Research 12 (2011) 2825–2830.

[49] P. Emami, A. Sahu, P. Graf, BuildingsBench: A large-scale dataset of 900K buildings and benchmark for short-term load forecasting, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track, 2023.

[50] J. Gao, J. Ploennigs, M. Bergés, A data-driven meta-data inference framework for building automation systems, in: Proceedings of the

2nd ACM International Conference on Embedded Systems for Energy-Efficient Built Environments (BuildSys), 2015, pp. 23–32. doi:10.1145/2821650.2821670.

[51] International Society of Automation, ANSI/ISA-18.2-2016: Management of alarm systems for the process industries, ISA, Research Triangle Park, NC (2016).

[52] M. Wetter, P. Ehrlich, A. Gautier, M. Grahovac, P. Haves, J. Hu, A. Prakash, D. Robin, K. Zhang, OpenBuildingControl: Digitizing the control delivery from building energy modeling to specification, implementation and formal verification, Energy 238 (2022) 121501. doi:10.1016/j.energy.2021.121501.

[53] W. Granzer, F. Praus, W. Kastner, Security in building automation systems, IEEE Transactions on Industrial Electronics 57 (11) (2010) 3622–3630. doi:10.1109/TIE.2009.2036033.

[54] M. Neukomm, V. Nubbe, R. Fares, Grid-interactive efficient buildings technical report series: Overview of research challenges and gaps, Tech. rep., U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Washington, DC (2019).

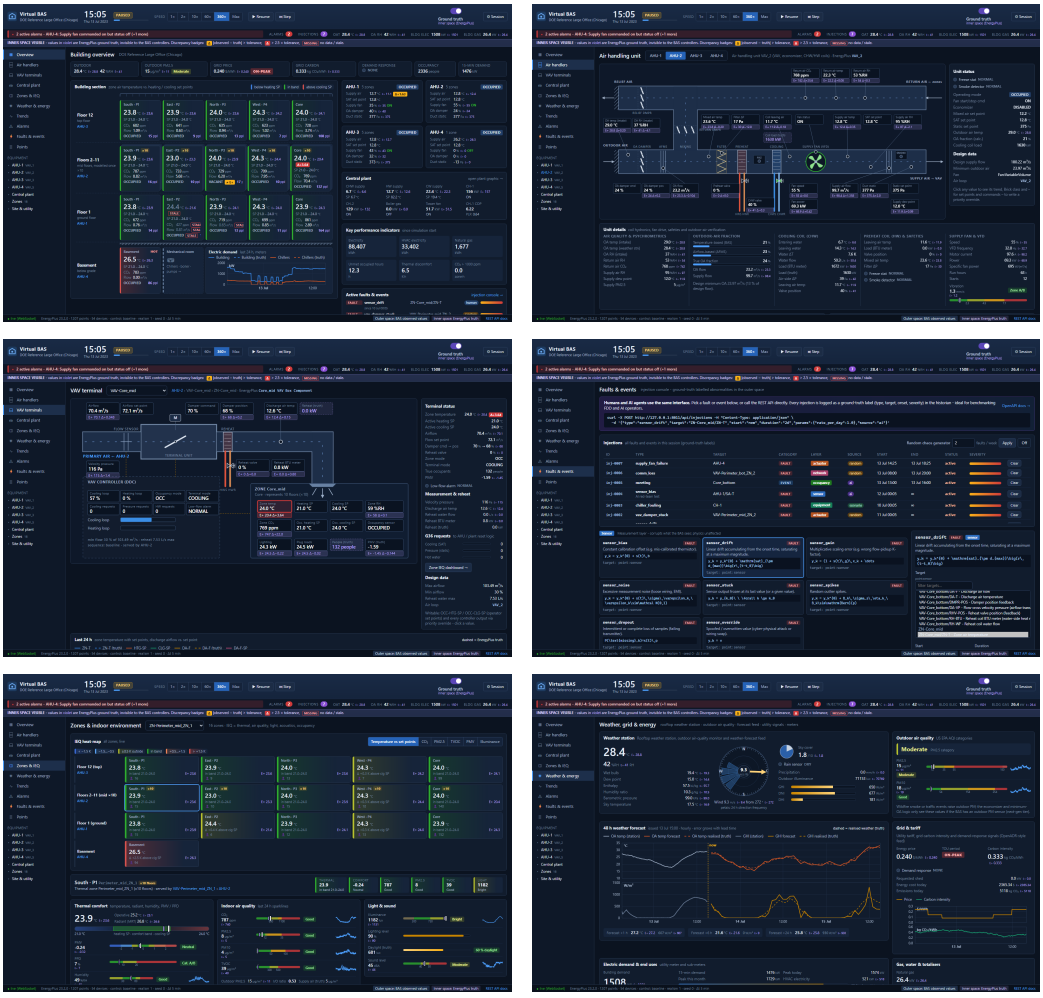


Figure 3: Web GUI of Virtual BAS with the ground-truth layer switched on: building overview (top left), AHU graphic (top right), VAV box with a drifting zone sensor (observed vs. EnergyPlus truth badge, middle left), fault and event injection console (middle right), zone indoor-environment page with a building heat-map (bottom left) and weather, forecast, grid and energy page (bottom right).

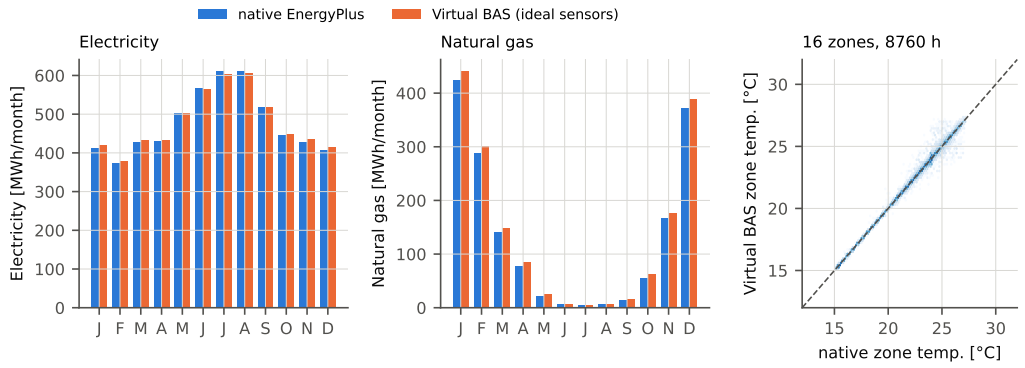


Figure 4: Experiment 1: monthly electricity and natural gas (left, centre) and hourly zone temperatures of all 16 zones (right) of native EnergyPlus and Virtual BAS with ideal sensors.



Figure 5: Experiment 2: drift of the zone temperature sensor of zone Core_mid. (a) Observed and true zone temperature; (b) discharge air flow of the VAV box with and without drift; (c) HVAC electric demand.

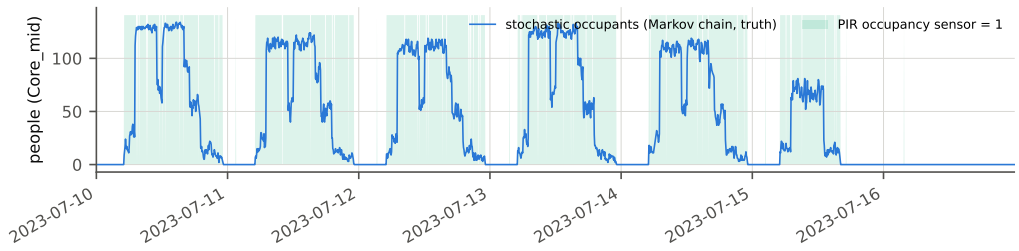


Figure 6: Stochastic occupancy of zone Core_mid (truth, written to EnergyPlus) and the corresponding PIR occupancy sensor seen by the BAS during one summer week.

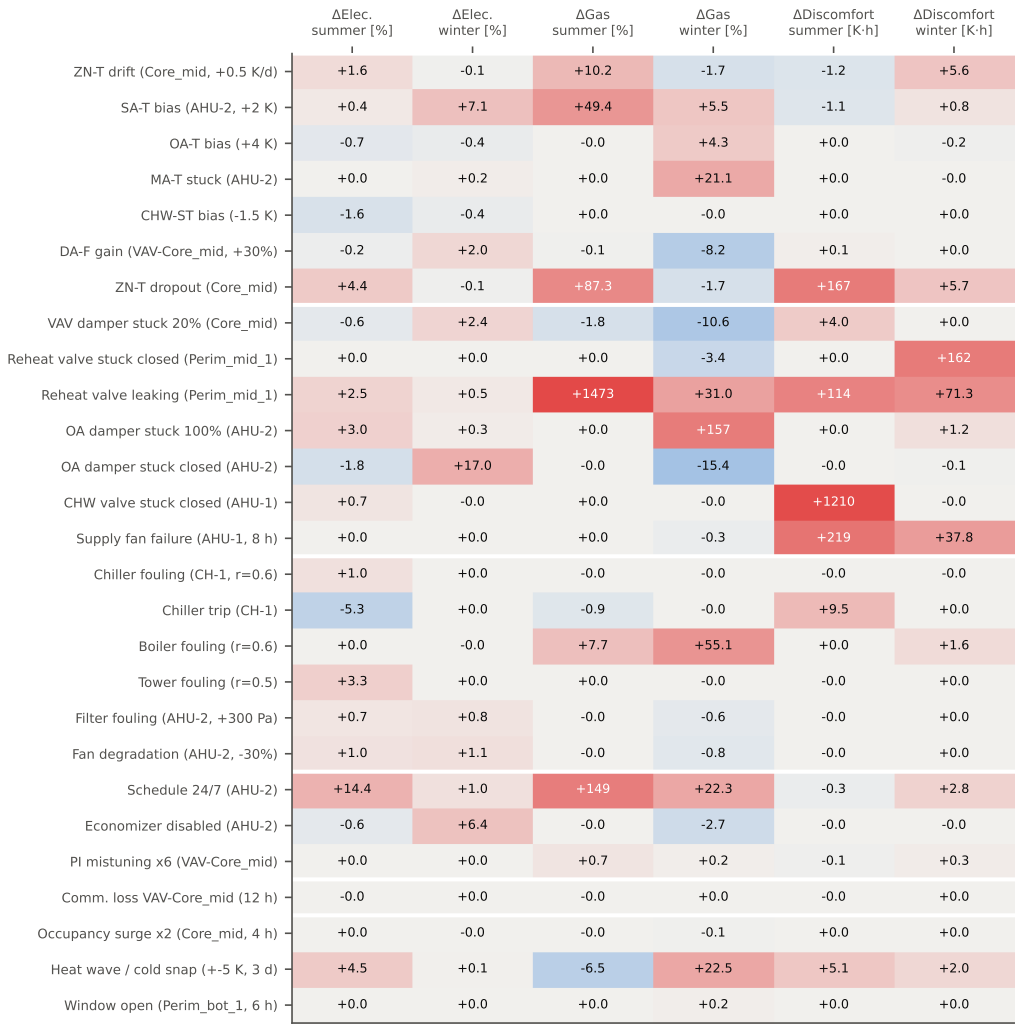


Figure 7: Experiment 3: change of electricity, natural gas and discomfort relative to the fault-free twin for 27 single-fault scenarios (one week each, summer and winter). Colour: signed logarithmic scale.

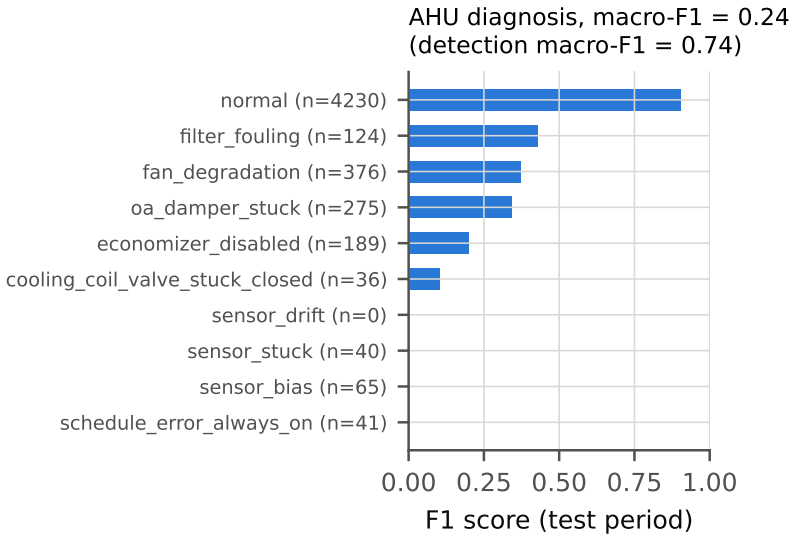


Figure 8: Experiment 4: per-class F1 score of the AHU fault diagnosis benchmark on the test period.

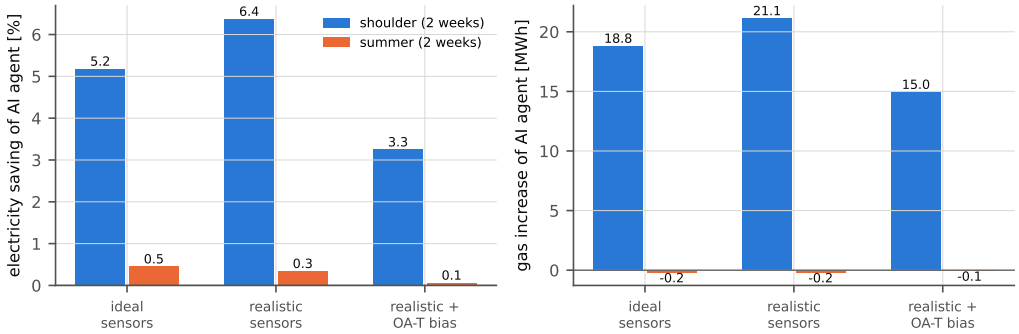


Figure 9: Experiment 5: electricity saving (left) and gas increase (right) of a supervisory AI controller relative to the baseline BAS program in three outer spaces (ideal sensors, realistic sensors, realistic sensors with a +4 K outdoor-air sensor bias).

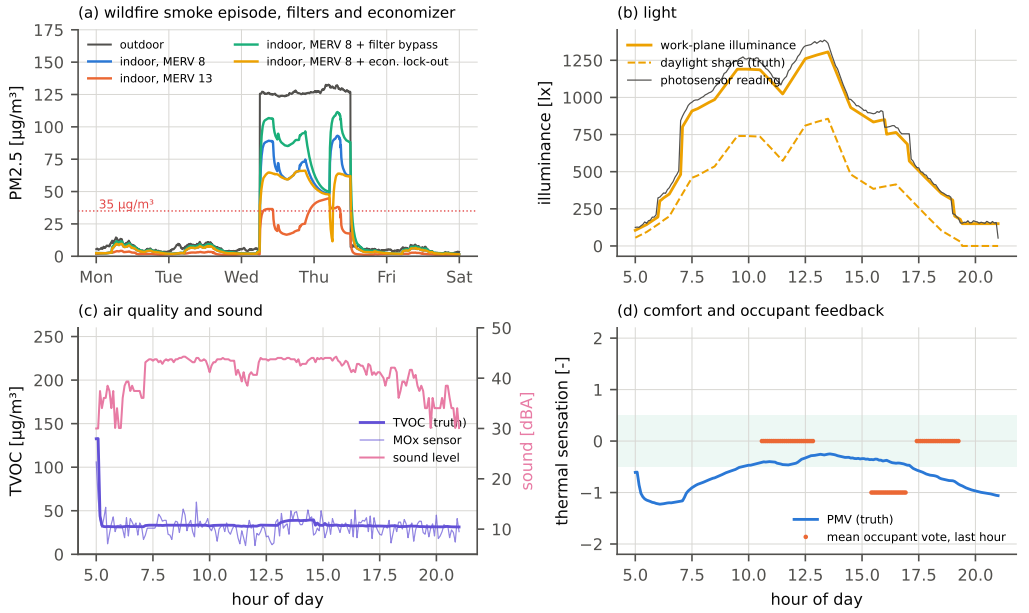


Figure 10: Experiment 6: (a) outdoor and mean indoor PM2.5 during a wildfire-smoke episode for four filter/control situations; (b)–(d) one summer day of illuminance, TVOC, sound and comfort points of a perimeter zone (truth and sensor readings).

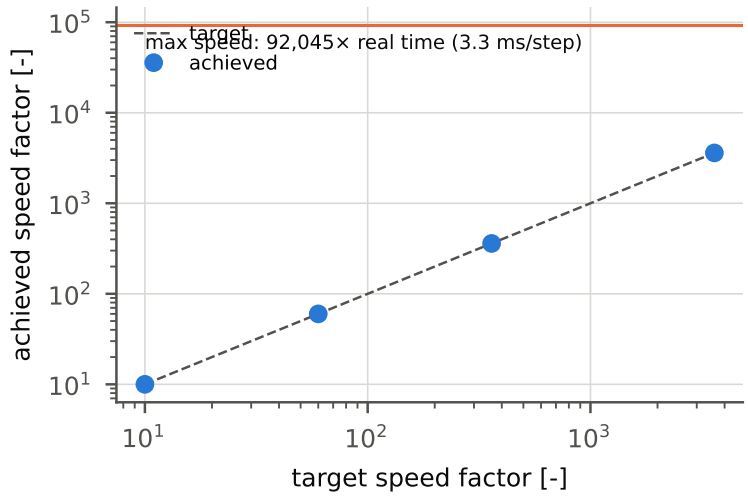


Figure 11: Experiment 7: target versus achieved speed factor and maximum throughput.